

MAT236 - Homework 1&2&3 (SENT ON 31.03.2026)**1 Questions****Q1. Solve** $w^n = z$, $n \in \mathbb{R}$.Let $z = re^{i\theta}$, $r = |z| > 0$, $\theta = \arg(z)$. Now, let $w = \rho e^{i\phi}$. Then

$$w^n = \rho^n e^{in\phi} = re^{i\theta}.$$

Equating:

$$\begin{aligned}\rho^n &= r \Rightarrow \rho = r^{1/n}, \\ n\phi &= \theta + 2\pi k, \quad k \in \mathbb{Z}.\end{aligned}$$

Thus

$$w_k = r^{1/n} e^{i\frac{\theta+2\pi k}{n}}, \quad k \in \mathbb{Z}.$$

Q2. Consider the complex equation $az^2 + bz + c = 0$. **When are the roots conjugate?**

We have

$$az^2 + bz + c = 0, \quad a, b, c \in \mathbb{R}.$$

Roots:

$$z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

The roots are conjugate iff

$$b^2 - 4ac < 0.$$

Q3. Solve $z^2 + \bar{z}^2 = -1$.Let $z = x + iy$, then $\bar{z} = x - iy$.

$$\begin{aligned}z^2 &= x^2 - y^2 + 2ixy \\ \bar{z}^2 &= x^2 - y^2 - 2ixy, \quad z^2 + \bar{z}^2 = 2(x^2 - y^2)\end{aligned}$$

$$2(x^2 - y^2) = -1 \Rightarrow x^2 - y^2 = -\frac{1}{2}.$$

Q4. Solve $\text{Arg}(z - i) = \frac{\pi}{2}$.Let $z = x + iy$, then

$$z - i = x + i(y - 1).$$

For $\text{Arg}(z - i) = \frac{\pi}{2}$

$$x = 0, \quad y - 1 > 0.$$

Thus

$$z = iy, \quad y > 1.$$